

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	P. Greeshma	Reg. No.:	192424418
Section / Batch:	2024-AI&DS	Date:	28-07-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

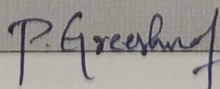
Selection Sort – Nearly Sorted Dataset :

Apply the Brute Force technique using Selection Sort on the dataset [5, 9, 11, 14, 8, 20]. Clearly interpret the problem and justify the use of Selection Sort for this dataset. Perform the sorting step-by-step and show intermediate results after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in different cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on a nearly sorted list.

Code of Conduct

I P. GREESHMA(192424418)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. GREESHMA



RUBRICS FOR CALCULATION

Name: _____

Signature: _____

Date: _____

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge

Course Coordinator

Head of Department

Name:

Signature: _____

Date: _____

Name:

Signature: _____

Date: _____

Name:

Signature: _____

Date: _____

Analytical problem.

P. Greeshma

192424418

* Selection Sort :-

- Selection Sort divides the array into a sorted and an unsorted parts
- It repeatedly selects the Minimum elements from the unsorted parts and move into the end of the sorted parts.

Algorithm:-

1. Start with the first position (index 0) at the current position
2. Search through the rest of the unsorted array to find the minimum element
3. Swap the smallest element found with the element at the current position
4. Move the current position one step forward
5. Repeat step 2 to 4 until only one element is remaining.

Pseudo code:-

- 1) set $i = 0$
- 2) Find Minimal $A[i \dots n-1]$
- 3) swap it with $A[i]$
- 4) increment i
- 5) Repeat steps 2 to 4 until $i = n-1$

Array [5, 9, 11, 14, 8, 20]

The Given dataset is [5, 9, 11, 14, 8, 20]

Pass 1: $\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \\ [5, & 9, & 11, & 14, & 8, & 20] \end{matrix}$

sorted position = 0

Min = 5 at index 0

No swap required.

pass 2 :- $\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \\ [5, & 9, & 11, & 14, & 8, & 20] \end{matrix}$

$\begin{matrix} \uparrow & \text{current} & \uparrow \\ & \text{position} & \text{Min} \end{matrix}$

sorted position = 1

Min = 8 at index 4

swap $9 \leftrightarrow 8$

Pass 3 :- $\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \\ [5, & 8, & 11, & 14, & 9, & 20] \end{matrix}$

$\begin{matrix} \leftarrow \text{sorted} \rightarrow \uparrow & \text{current} & \uparrow \\ & \text{position} & \text{Min} \end{matrix}$

sorted position = 2

Min = 9 at index 4

swap $11 \leftrightarrow 9$

pass 4 :- $\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \\ [5, & 8, & 9, & 14, & 11, & 20] \end{matrix}$

$\begin{matrix} \leftarrow \text{sorted} \rightarrow \uparrow & \text{current} & \uparrow \\ & \text{position} & \text{Min} \end{matrix}$

sorted position = 3

Min = 11 at index 4

swap $11 \leftrightarrow 14$

pass 5 :- [5, 8, 9, 11, 14, 20]

← Sorted →

Total comparisons :-

For selection sorts.

- pass 1 → 5
- pass 2 → 4
- pass 3 → 3
- pass 4 → 2
- pass 5 → 1

$$\text{Total comparisons} = 5 + 4 + 3 + 2 + 1$$

$$= 15$$

Total swaps :-

- pass 1 → 0
- pass 2 → 1
- Pass 3 → 1
- pass 4 → 1
- pass 5 → 0

$$\text{Total swaps} = 0 + 1 + 1 + 1 + 0$$

$$= 3$$

Case:- Average Case

Time complexity:- $O(n^2)$ because Selection Sort always scans the unsorted positions of the array regardless of its initial order.